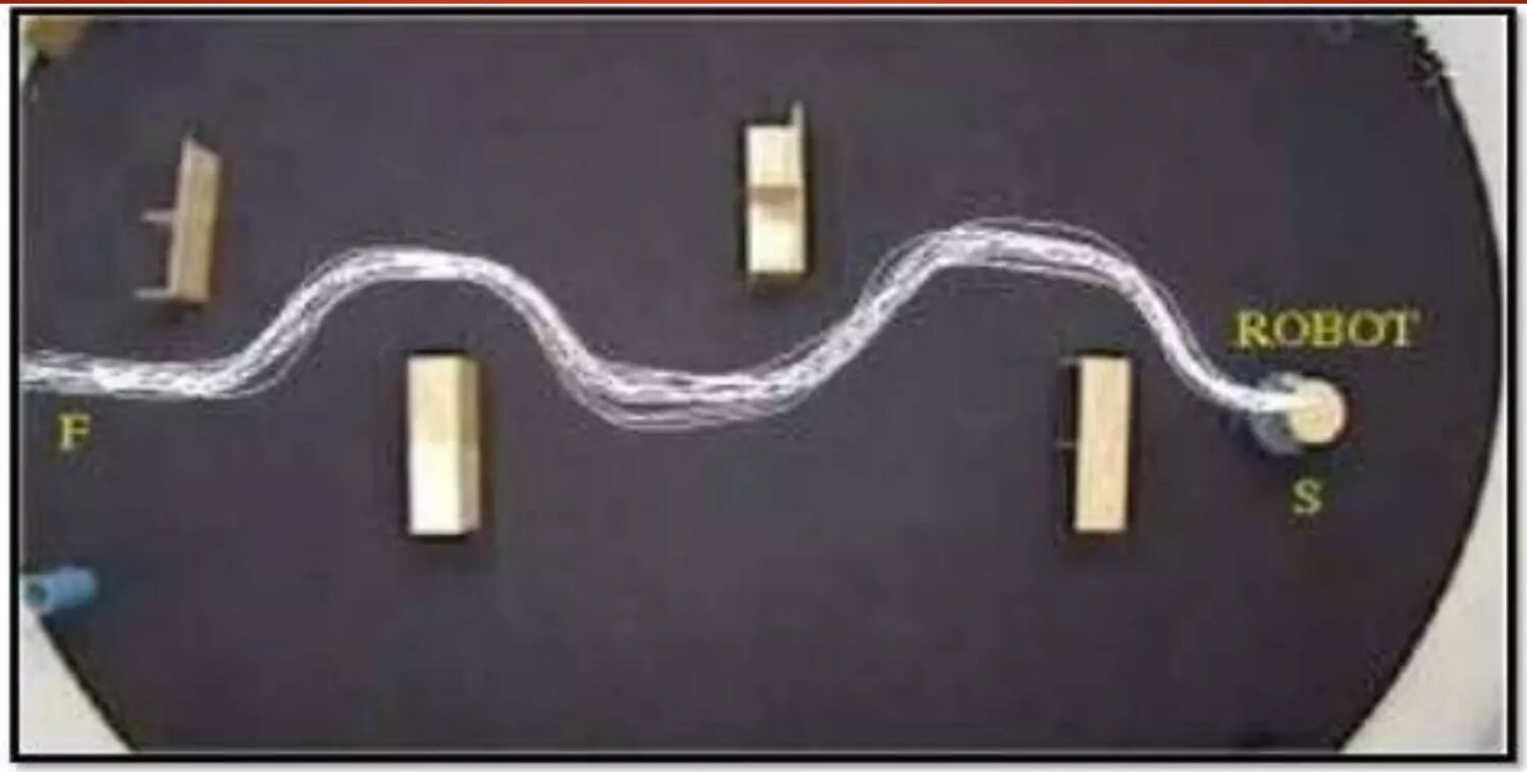
OBSTACLE AVOIDING ROBOT

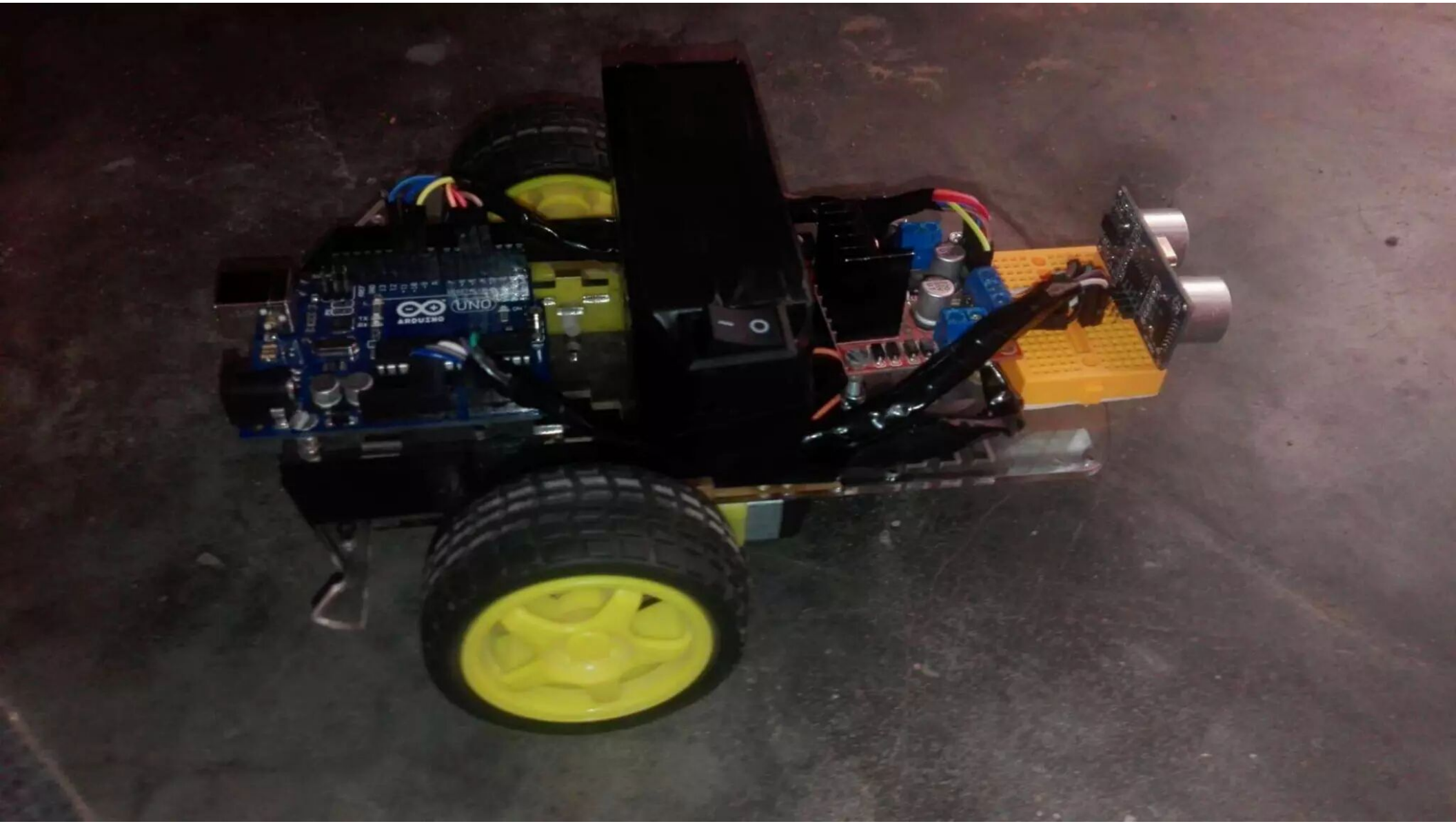
What is Obstacle Avoidance ??

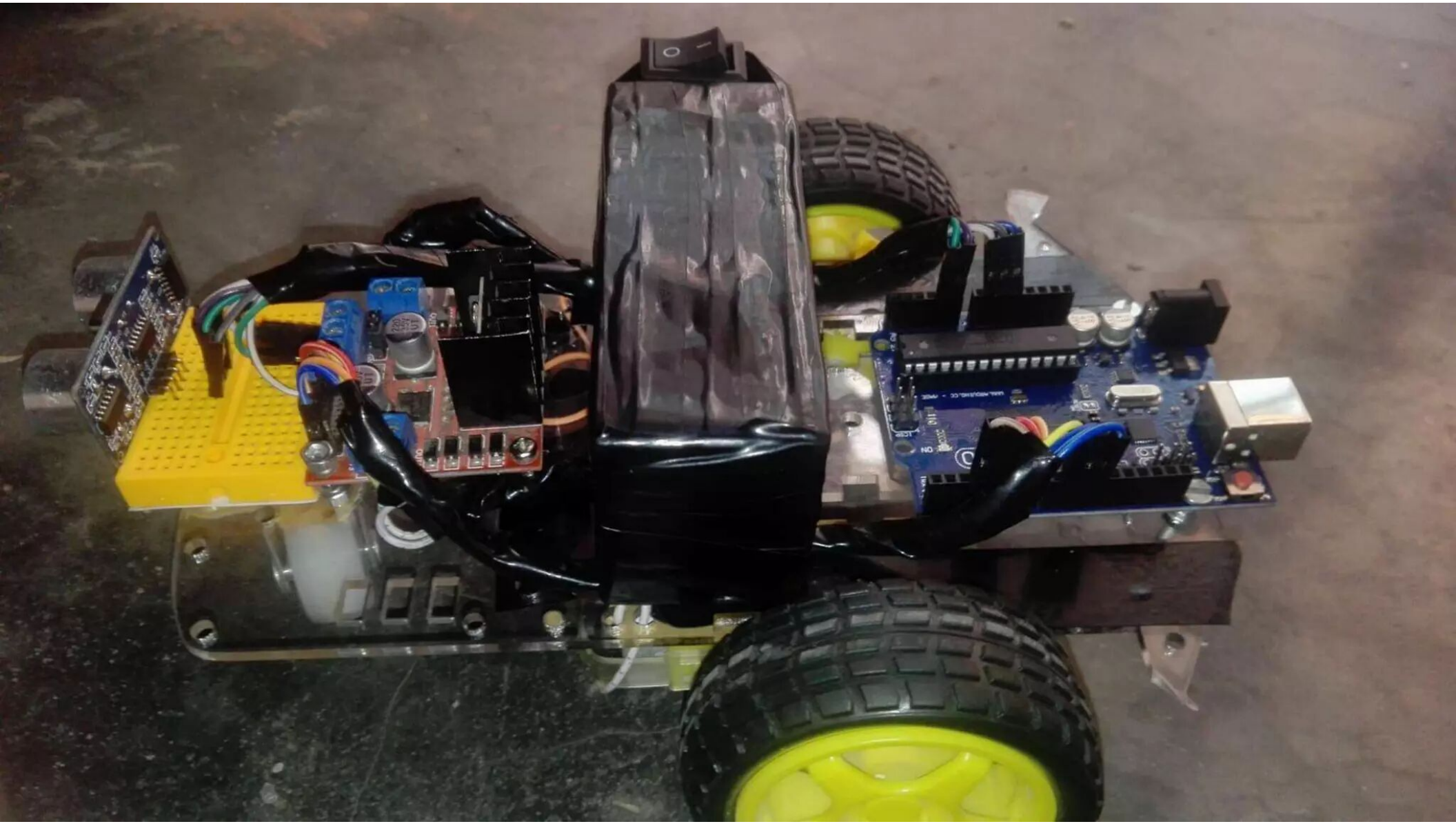


A Path Describing The Purpose Of This Type Robot

Our Robot

- The obstacle detection is primary requirement of this autonomous robot
- In this project our robot senses any obstacle in its path, avoids it and resume its running
- Involves the pre-computation of an obstacle-free path which a controller guides the robot


UNO
ARDUINO

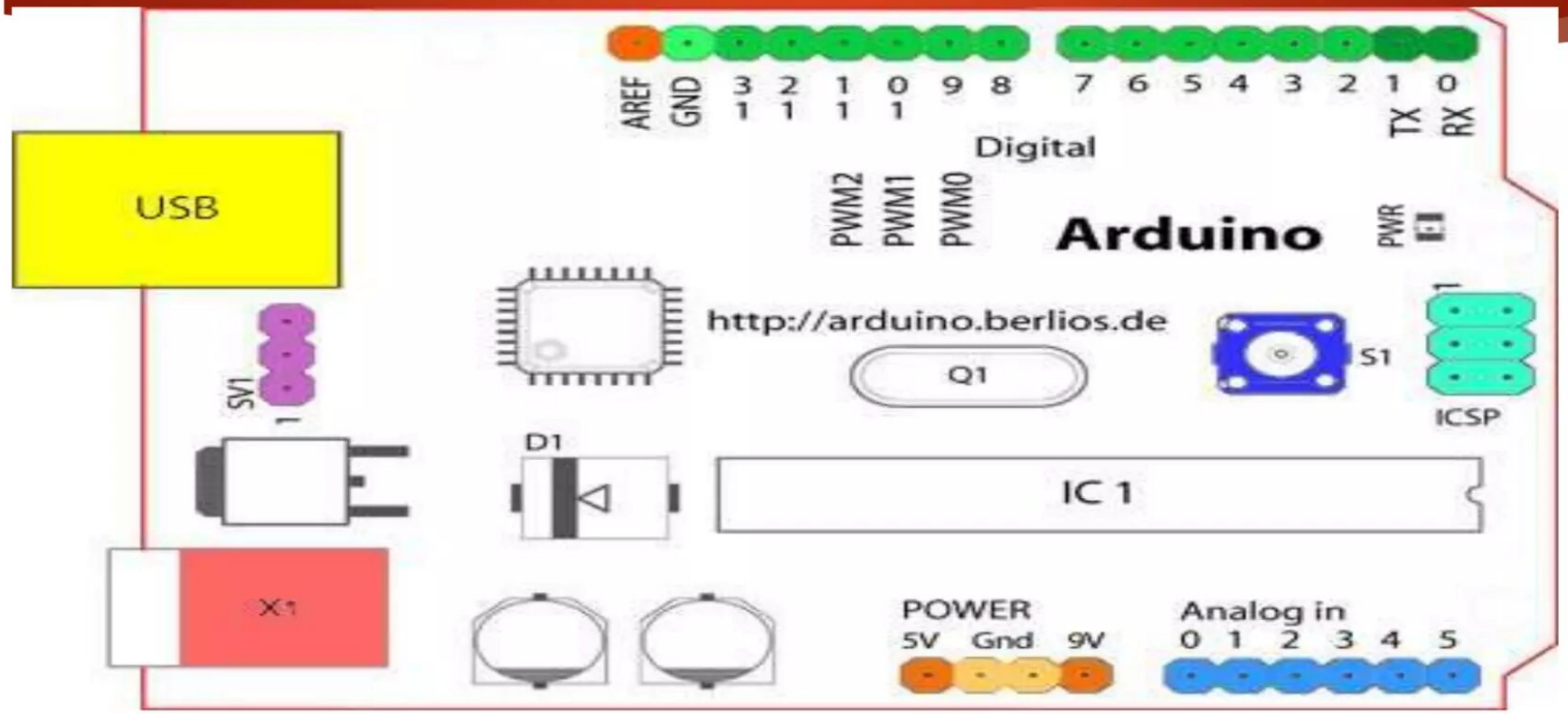


Hardware

- UNO Arduino
- U.R Sensor
- DC motor Driver
- Connecting wires
- USB programmer

1- UNO Arduino:

- Arduino is a software company, project, and user community that designs and manufactures computer open-source hardware, open-source software, and microcontroller-based kits for building digital devices and interactive objects that can sense and control physical devices.


AREF
GND
3
2
1
0
9
8
7
6
5
4
3
2
1
0
TX
RX
Digital
PWM2
PWM1
PWM0
Arduino
PWR
USB
http://arduino.berlios.de
Q1
S1
ICSP
SV1
D1
IC 1
X1
POWER
5V
Gnd
9V
Analog in
0
1
2
3
4
5

Sensors

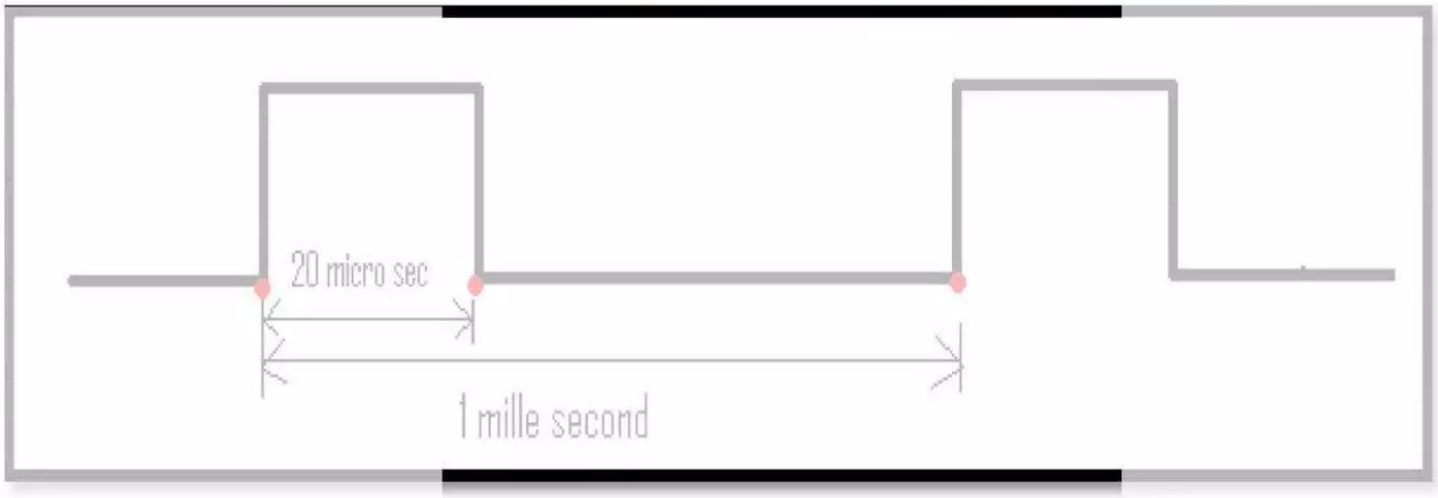
- Some sensing devices used for obstacle detection like bump sensor, infrared sensor, ultrasonic sensor etc.

- Ultrasonic sensor is most suitable for obstacle detection because of

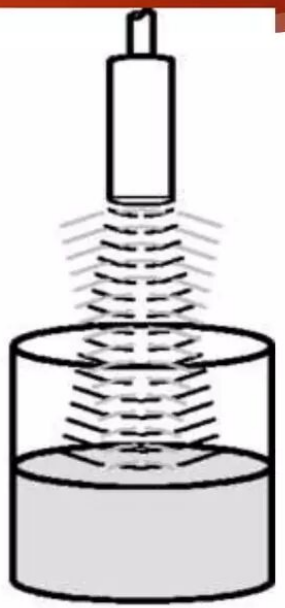
- **LOW COST**
- **HIGH RANGING CAPABILITY**

2- Ultrasonic Sensor

- Emit a sound pulse that reflects off of objects entering the wave field
- The reflected sound, or "echo" is then received by the sensor
- Detection of the sound generates an output signal for use by an actuator, controller, or computer
- The output signal can be analog or digital.
- The time for an ultrasonic sensor's beam to strike the target and return is directly proportional to the distance to the object

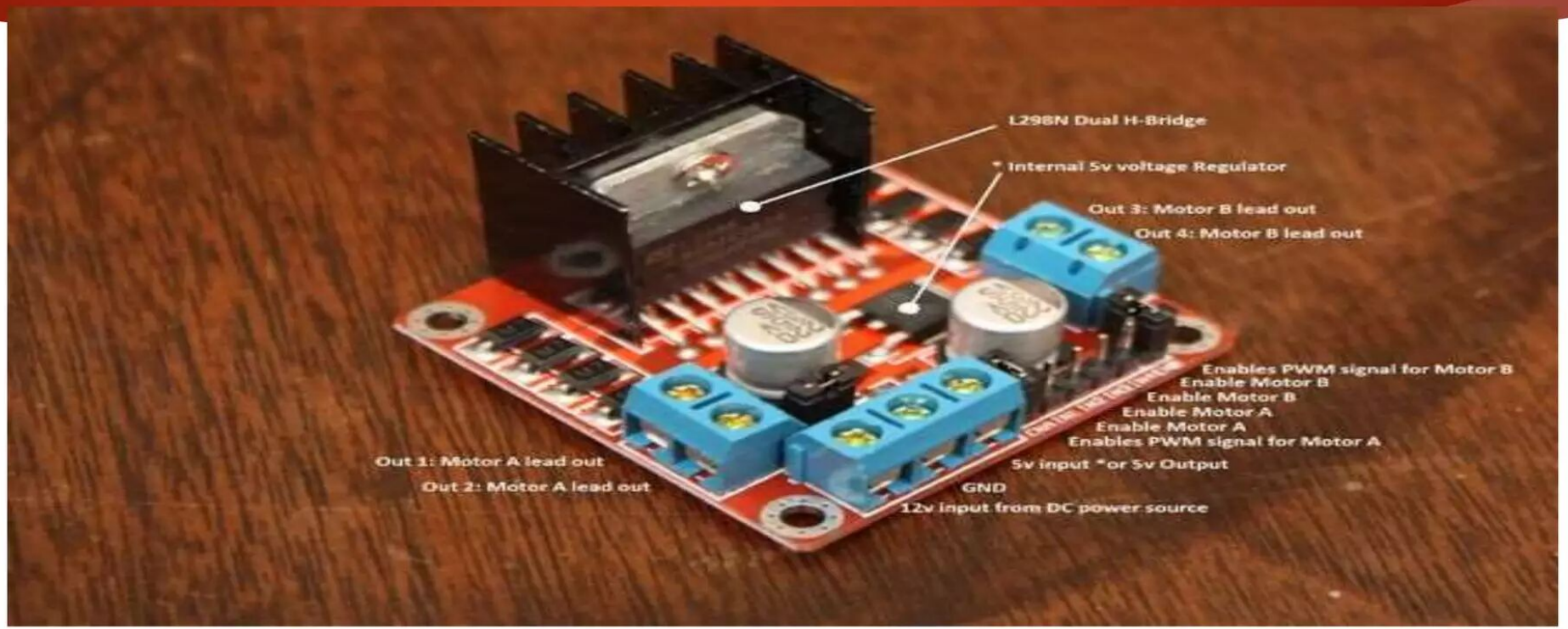

20 micro sec
1 mille second


Sensor
Target



3- DC Motor Driver (L298N)

- L298N is a typical Motor driver or Motor Driver IC which allows DC motor to drive on either direction.
- L298N is a 16-pin IC which can control a set of two DC motors simultaneously in any direction.
- It means that you can control two DC motor with a single L298N IC.


L298N Dual H-Bridge
* Internal 5v voltage Regulator
Out 3: Motor B lead out
Out 4: Motor B lead out
Enables PWM signal for Motor B
Enable Motor B
Enable Motor B
Enable Motor A
Enable Motor A
Enables PWM signal for Motor A
5v input *or 5v Output
GND
12v input from DC power source
Out 1: Motor A lead out
Out 2: Motor A lead out

Specifications:

- Double H bridge Drive Chip: *L298N*
- Logical voltage: *5V Drive voltage: 5V-35V*
- Logical current: *0-36mA Drive current: 2A (MAX single bridge)*
- Max power: *25W*
- Dimensions: *43 x 43 x 26mm*
- Weight: *26g*

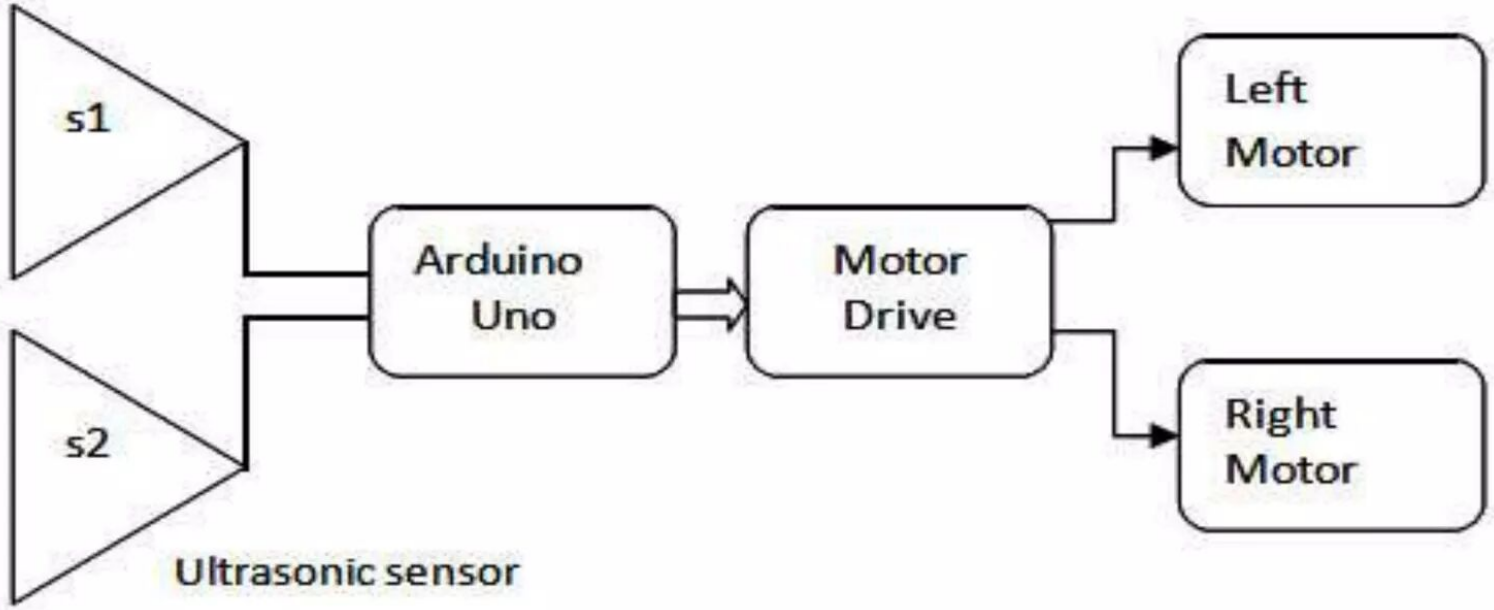
4- Connecting wires :

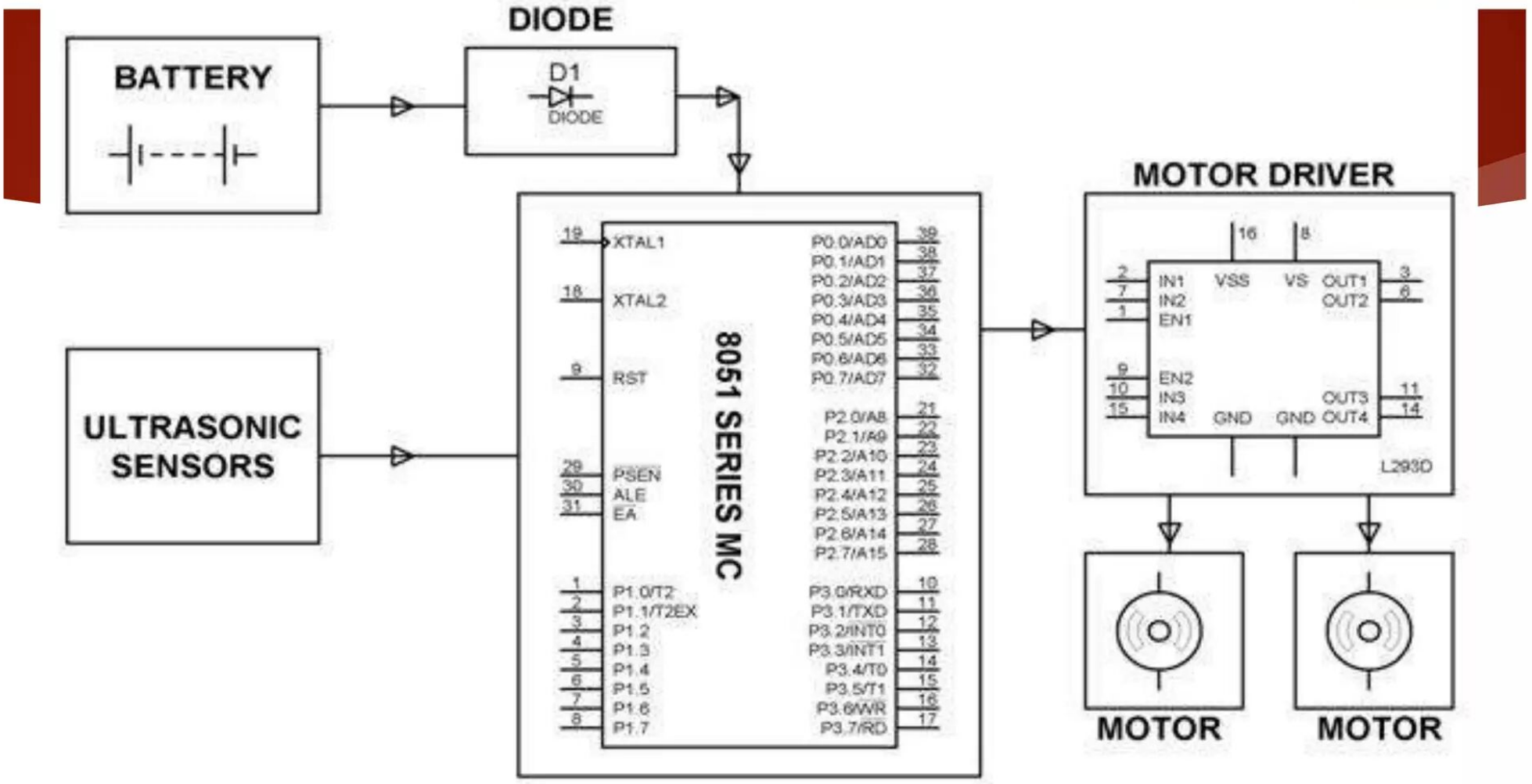
We are using these wires for making the connections.

5- USB Programmer:

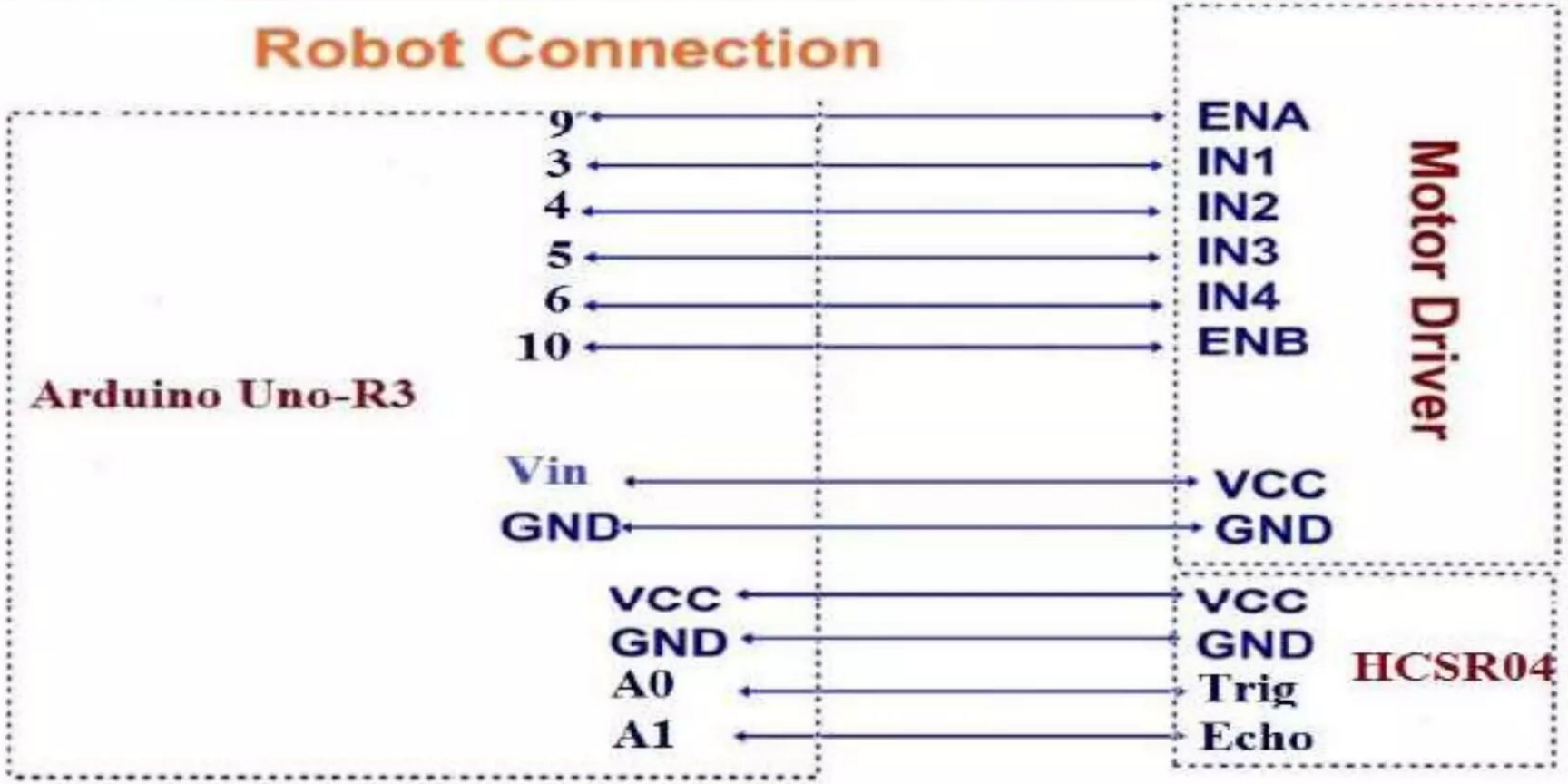
USB Programmer is basically use for uploading the code from computer to Arduino.

Block Diagram




DIODE
BATTERY
D1
DIODE
MOTOR DRIVER
ULTRASONIC SENSORS
8051 SERIES MC
XTAL1
XTAL2
RST
PSEN
ALE
EA
P1.0/T2
P1.1/T2EX
P1.2
P1.3
P1.4
P1.5
P1.6
P1.7
P0.0/AD0
P0.1/AD1
P0.2/AD2
P0.3/AD3
P0.4/AD4
P0.5/AD5
P0.6/AD6
P0.7/AD7
P2.0/A8
P2.1/A9
P2.2/A10
P2.3/A11
P2.4/A12
P2.5/A13
P2.6/A14
P2.7/A15
P3.0/RXD
P3.1/TXD
P3.2/INT0
P3.3/INT1
P3.4/T0
P3.5/T1
P3.6/WR
P3.7/RD
IN1
IN2
EN1
EN2
IN3
IN4
VSS
VS
GND
GND
OUT1
OUT2
OUT3
OUT4
L293D
MOTOR
MOTOR

Circuit Diagram



Programe Coding

- #define trigPin
- #define echoPin
- #define MotorA_IN1
- #define MotorA_IN2
- #define MotorB_IN3
- #define MotorB_IN4
- #define MotorA_PWM
- #define MotorB_PWM

```
void setup()
  {
  pinMode(MotorA_IN1, OUTPUT);
  pinMode(MotorA_IN2, OUTPUT);
  pinMode(MotorB_IN3, OUTPUT);
  pinMode(MotorB_IN4, OUTPUT);
  pinMode(MotorA_PWM, OUTPUT);
  pinMode(MotorB_PWM, OUTPUT);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
  }
```

```
float search(void)
 {
   float duration = 0.00;
   float CM = 0.00;
     digitalWrite(trigPin, LOW);
     delayMicroseconds(2);

     //Send 10us High Pulse to Ultra-Sonic Sonar Sensor "trigPin"
     digitalWrite(trigPin, HIGH);
     delayMicroseconds(10);
     digitalWrite(trigPin, LOW);
   duration = pulseIn(echoPin, HIGH);
   CM = (duration / 58.82);
  return CM;
 }
```

```
void RobotForward()
  {
  digitalWrite(MotorA_IN1, HIGH);
  digitalWrite(MotorA_IN2, LOW);
  digitalWrite(MotorB_IN3, HIGH);
  digitalWrite(MotorB_IN4, LOW);
  }
void RobotBackward()
  {
  digitalWrite(MotorA_IN1, LOW);
  digitalWrite(MotorA_IN2, HIGH);
  digitalWrite(MotorB_IN3, LOW);
  digitalWrite(MotorB_IN4, HIGH);
  }
```

```
void RobotLeft()
  {
  digitalWrite(MotorA_IN1, LOW);
  digitalWrite(MotorA_IN2, HIGH);
  digitalWrite(MotorB_IN3, HIGH);
  digitalWrite(MotorB_IN4, LOW);
  }
void RobotRight()
  {
  digitalWrite(MotorA_IN1, HIGH);
  digitalWrite(MotorA_IN2, LOW);
  digitalWrite(MotorB_IN3, LOW);
  digitalWrite(MotorB_IN4, HIGH);
 }
```

```
void RobotStop()
  {
  digitalWrite(MotorA_IN1, LOW);
  digitalWrite(MotorA_IN2, LOW);
  digitalWrite(MotorB_IN3, LOW);
  digitalWrite(MotorB_IN4, LOW);
  }
void loop()
  {
     float distance = 0.00;
     float RobotSpeed = 0.00;
     float RightDistance = 0.00;
     float LeftDistance = 0.00
    distance = search();
```

```
if((distance <= 40))  .
    {
      RobotSpeed = 50;
         analogWrite(MotorA_PWM, RobotSpeed);
         analogWrite(MotorB_PWM, RobotSpeed);
      RobotStop();
      delay(10);
      RobotBackward();
      delay(400);
      RobotStop();
      delay(10);
      RobotRight();
      delay(500);
      RightDistance = search();
      delay(10);
      RobotLeft();
      delay(900);
```

```
LeftDistance = search();
      delay(10);
       if(LeftDistance >= RightDistance)
        {
          RobotForward();
        }
        else
        {
          RobotRight();
          delay(800);
          RobotStop();
          delay(10);
          RobotForward();
        }
    }
```

```
else if((distance > 40) && (distance <= 75))
    {
     RobotSpeed = 150;
          analogWrite(MotorA_PWM, RobotSpeed);
          analogWrite(MotorB_PWM, RobotSpeed);
     RobotForward();
    }
  else
    {
        RobotSpeed = 200;
          analogWrite(MotorA_PWM, RobotSpeed);
          analogWrite(MotorB_PWM, RobotSpeed);
       RobotForward();

```

Applications

- Automated Cars (Google Car)
- Toys
- Military
- Mines